

Software Requirements Specification

IBA Campus Facility Booking System

1. Introduction (Problem Statement, Objectives and Stake Holders)

1.1 Problem Statement (Real-World Context)

The Institute of Business Administration relies on shared campus facilities such as classrooms, seminar halls, computer labs, and meeting rooms to support academic programs, administrative activities, and extracurricular events. Currently, the booking of these facilities is often handled through informal or semi-manual methods such as emails, physical registers, or verbal coordination with administrative staff.

These methods, however, present several challenges. They lack real-time visibility into facility availability, do not consistently abide by booking rules, and frequently result in scheduling conflicts, or double bookings. Administrators face difficulties in tracking requests, managing approvals, and maintaining accurate records of facility usage. Students and faculty may experience delays and uncertainty in the booking process, which can affect teaching, learning, and event organization.

There is a clear need for a centralized system that allows users at IBA to request facility bookings, ensures constraints, and enables administrators to efficiently manage approvals and facility availability. A Campus Facility Booking System addresses these challenges by providing a structured and reliable solution for managing campus resources effectively.

1.2 Objectives

The primary objective of the Campus Facility Booking System is to streamline the process of reserving campus facilities and provide a centralized platform which is easy to navigate for all users.

Specific objective (success of the project)s:

- Allow students and faculty to request bookings for classrooms, seminar rooms, auditoriums, computer labs and meeting rooms in an easy and structured manner.
- Ensure the facility is available and booking rules are being applied and followed.
- Provide administrators with a standardized approval procedure for reviewing, accepting, or rejecting booking requests.
- Show real-time facility availability and booking status.
- Maintain accurate records of users, facilities and bookings for reporting and administrative purposes.

Success criteria (how it will be measured):

- Users can successfully submit booking requests without conflicts or double bookings.
- Administrators can process booking approvals efficiently, following all booking rules and constraints.
- All delays caused by manual booking methods are reduced, through an easy request and approval system.
- Facility availability during a specific date and time is visible to all relevant users.

- The system reliably supports multiple user roles, such as students/faculty, program office and admin, with minimal manual intervention.
- Accurate records of bookings, users and facility usage are maintained for future reference.

1.3 Stake Holders

The key stakeholders for the IBA Campus Facility Booking System are the direct users—students and faculty who will request bookings, Program Office staff who will approve them, and system administrators who will manage the platform.

2. System Features

2.1 Authentication (Student / PO)

2.1.1 Student enters ERP and password

2.1.2 Program Office (PO) enters username and password

2.1.3 System validates credentials and redirects to dashboard

Description and Priority

- **Description:** Allows students and program office members to securely log in to the system using their credentials.
- **Priority:** High

Stimulus/Response Sequences

1. Student enters ERP and password → System validates → Redirects to student dashboard if valid; shows error if invalid.
2. PO enters username and password → System validates → Redirects to PO dashboard if valid; shows error if invalid.

Functional Requirements

- **REQ-1:** System must provide a login form for students with ERP and password fields.
- **REQ-2:** System must provide a login form for program office members with username and password fields.
- **REQ-3:** System must validate credentials against stored user data.
- **REQ-4:** On valid login, the system must redirect the user to their respective dashboard.
- **REQ-5:** On invalid login, the system must display an appropriate error message without revealing sensitive information.
- **REQ-6:** System should prevent unauthorized access to dashboards without successful login.

2.2 Request Room Booking (Student)

2.2.1 Student selects a building

2.2.2 Student selects a room within the building

2.2.3 Student selects a date and time duration (9 AM – 6 PM, any date in current year)

2.2.4 Student enters purpose and submits request

Description and Priority

- **Description:** Allows a student to request a room for a specific date and time, specifying the purpose of booking. The system checks availability and sends the request to the Program Office if the room is free.
- **Priority:** High

Stimulus/Response Sequences

1. Student selects a building → System displays available rooms in that building.
2. Student selects a room → System shows booking calendar for that room.
3. Student selects a date and time duration → System checks availability.
 - If the room is booked or disabled → System shows an error message.
 - If room is free → System allows submission of request.
4. Student enters purpose and clicks **Request** → System forwards request to Program Office for approval and confirms submission to student.

Functional Requirements

- **REQ-1:** System must display a list of buildings for selection.
- **REQ-2:** System must display available rooms within the selected building.
- **REQ-3:** System must allow selection of a date (any date in current year) and a time duration between 9 AM and 6 PM.
- **REQ-4:** System must prevent booking if the room is already booked or disabled at the selected time.
- **REQ-5:** System must allow the student to enter a purpose for the booking.
- **REQ-6:** On successful request, system must forward the request to Program Office and confirm submission to the student.
- **REQ-7:** On conflict (room booked/disabled), the system must show an appropriate error message.
- **REQ-8:** System must maintain a booking record for future reference.

2.3 Accept/Reject Room Booking (PO)

2.3.1 PO views pending room booking requests

2.3.2 PO selects a request and approves or rejects it

2.3.3 System updates status and notifies student

Description and Priority

- **Description:** Allows Program Office members to manage room booking requests from students.
- **Priority:** High

Stimulus/Response Sequences

1. PO logs in → System displays pending booking requests.
2. PO selects a request → PO clicks **Approve** or **Reject** → System updates booking status.
3. The system notifies students of the decision.

Functional Requirements

- **REQ-1:** System must list all pending booking requests.
- **REQ-2:** System must allow PO to approve or reject a request.
- **REQ-3:** System must update room schedule if approved.
- **REQ-4:** System must notify the student of approval or rejection.
- **REQ-5:** System must prevent double booking if approved.

2.4 Add Student (Admin)

2.4.1 Admin enters student details

2.4.2 Admin submits the form

2.4.3 System creates student account

Description and Priority

- **Description:** Allows admins to add new students to the system.
- **Priority:** High

Stimulus/Response Sequences

1. Admin fills in student details → System validates inputs → System creates account or shows errors.

Functional Requirements

- **REQ-1:** System must accept student details (ERP, name, email, etc.)
- **REQ-2:** System must validate unique ERP.
- **REQ-3:** System must create a student account and notify admin of success.
- **REQ-4:** System must show error if input is invalid or ERP exists.

2.5 Add Program Office Member (Admin)

2.5.1 Admin enters PO details

2.5.2 Admin submits form

2.5.3 System creates PO account

Description and Priority

- **Description:** Allows admins to add new program office members.
- **Priority:** High

Functional Requirements

- **REQ-1:** System must accept PO details (name, username, email, password).
- **REQ-2:** System must ensure username is unique.
- **REQ-3:** System must create PO account and notify admin.
- **REQ-4:** System must show error if input invalid or username exists.

2.6 Disable Room (Admin)

- 2.6.1 Admin selects a room
- 2.6.2 Admin disables the room for a certain amount of time
- 2.6.3 System prevents bookings for that room

Description and Priority

- **Description:** Allows admins to temporarily or permanently disable a room for maintenance or other reasons for a certain amount of time.
- **Priority:** Medium

Functional Requirements

- **REQ-1:** System must allow admin to select a room and disable it.
- **REQ-2:** System must prevent future bookings for disabled rooms.
- **REQ-3:** System must show a status of “disabled” for the room.

2.7 Add Room (Admin)

- 2.7.1 Admin enters room details (name, building, capacity)
- 2.7.2 Admin submits form
- 2.7.3 System adds the room to the database

Description and Priority

- **Description:** Allows admins to add new rooms to the system for booking.
- **Priority:** Medium

Functional Requirements

- **REQ-1:** System must accept room details (name, building, capacity).
- **REQ-2:** System must validate unique room names within a building.
- **REQ-3:** System must add the room to the database.
- **REQ-4:** System must show confirmation of successful addition.

2.8 Add Building(Admin)

- 2.8.1 Admin enters building details
- 2.8.2 Admin submits form
- 2.8.3 System adds the building to the database

Description and Priority

- **Description:** Allows admins to add new buildings to the system.
- **Priority:** Medium

Functional Requirements

- **REQ-1:** System must accept building details (name, location, etc.)
- **REQ-2:** System must validate unique building names.

- **REQ-3:** System must add building to the database.
- **REQ-4:** System must show confirmation of successful addition.

2.9 Cancel Booking (Student / PO / Admin)

2.9.1 User selects a booking

2.9.2 User cancels booking

2.9.3 System updates status

Description and Priority

- **Description:** Allows users to cancel a booking before the booking start time.
- **Priority:** Medium

Stimulus/Response Sequences

User clicks “Cancel” → System validates → Booking status updated.

Functional Requirements

- **REQ-1:** System must allow cancellation before booking start time.
- **REQ-2:** System must update status to “Cancelled”.
- **REQ-3:** System must free the room slot after cancellation.

Implementation Coverage Note

All system features and requirements described above will be implemented exactly as specified in the IBA Campus Facility Booking System.